User Engagement Analysis & Insights

Synaptix Digital Solutions Ltd

Technical Project Report

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1. Introduction

In today's digital economy, organizations increasingly rely on user engagement analytics to understand how customers interact with digital platforms. Measuring engagement allows businesses to identify behavioural patterns, evaluate user experience, and develop strategies that improve user retention and overall platform performance. This project analyses user engagement data for Synaptix Digital Solutions in order to uncover patterns in user behaviour and identify opportunities to improve engagement across the platform. An interactive Power BI dashboard was developed to explore user engagement metrics across multiple dimensions including demographic groups, device types, geographic locations, and usage behavior.

2. Project Objectives

The objective of this project was to perform an in-depth analysis of user engagement data to identify behavioral patterns and generate actionable insights that improve platform performance and user retention. Key objectives included:

- Analyze user engagement behavior using key metrics.

- Identify high and low engagement user segments.

- Evaluate engagement patterns across age groups, device types and locations.

- Identify users at risk of churn.

- Develop an interactive dashboard to visualise engagement insights.

3. Tools and Technologies Used

Excel – Data cleaning and preprocessing

SQL – Data extraction and querying

Power BI – Data analysis and interactive dashboard development

PowerPoint – Communication of analytical insights

4. Dataset Overview

The dataset used in this project contains user activity and engagement information collected from the Synaptix digital platform. Key variables analyzed include User ID, Age Group, Gender, Device Type, Location, Session Duration, Number of Sessions, App Usage Hours, Engagement Score, Usage Segment and Risk Level.

5. Power BI Dashboard

The following dashboards present key engagement metrics and user behavior patterns identified during the analysis

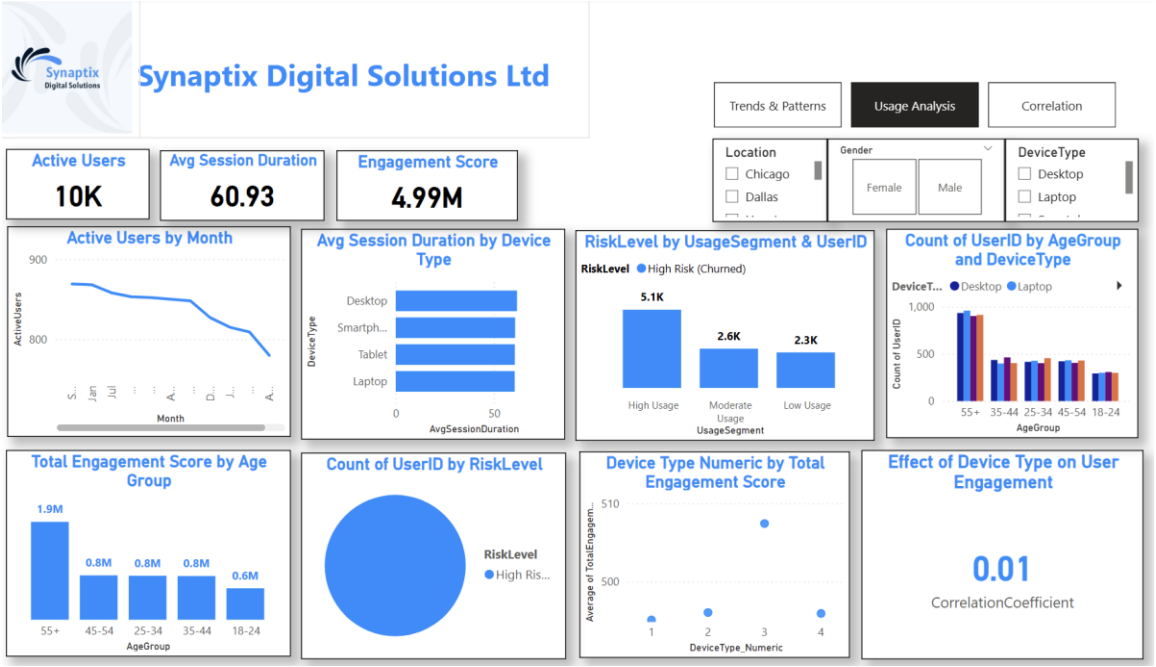

Figure 1 User engagement analysis shows how engagement scores vary across age groups, device types, and geographic locations.

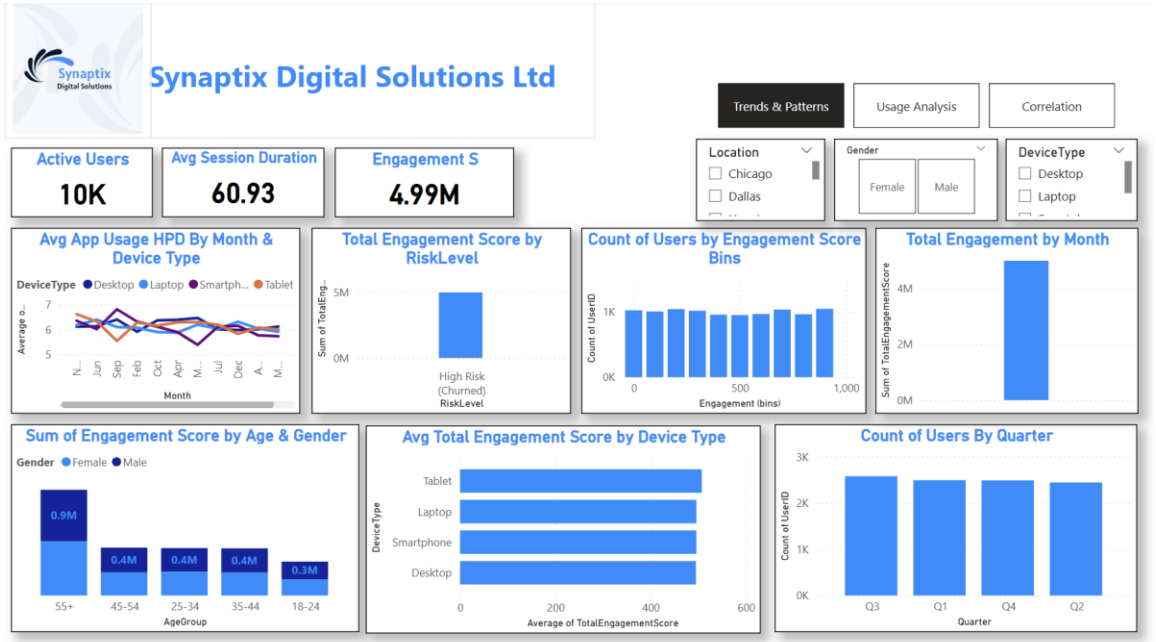

Figure 2 User engagement trends and patterns showing engagement score distribution across device types, locations, and age groups.

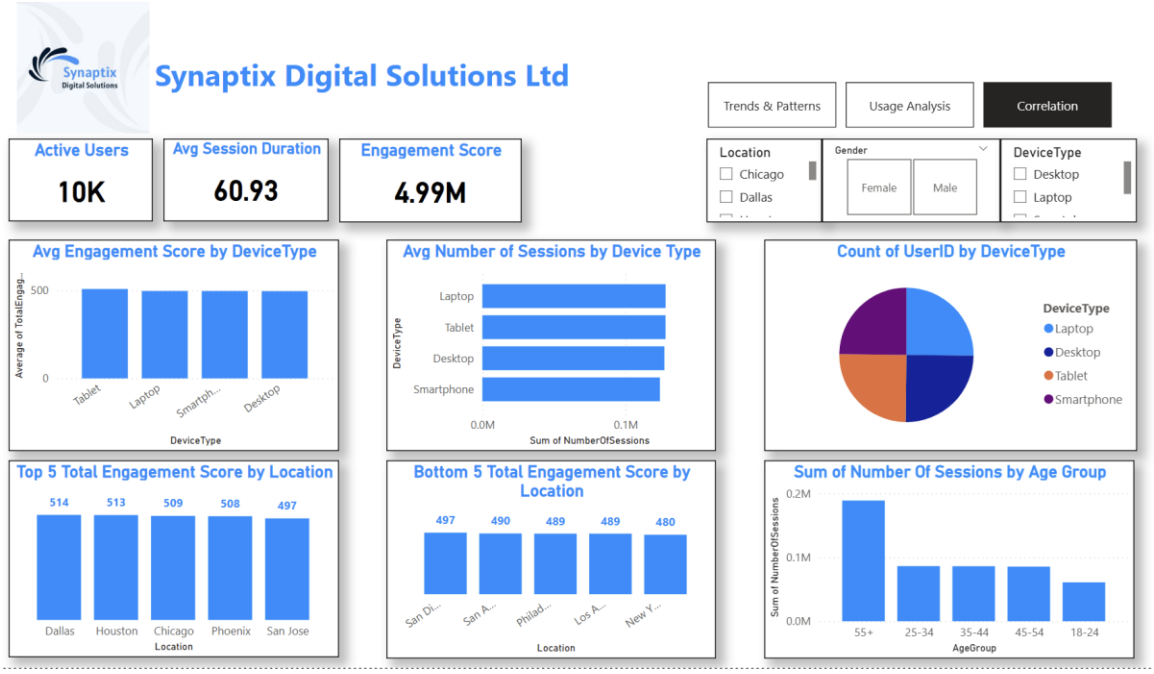

Figure 3 **Summary dashboard highlighting key engagement insights, including high-performing user segments, churn risk indicators, and engagement distribution across regions and devices**

6. Key Metrics Analyzed

Active Users: 10,000

Average Session Duration: 60.93 minutes

Total Engagement Score: 4.99 million

7. Key Analytical Insights

Users aged 55+ demonstrate the highest engagement levels. Users aged 45–54 show the second-highest engagement. Tablet and laptop users tend to have higher engagement scores. Phoenix, Chicago and Houston demonstrate higher engagement levels. Users with lower usage hours and fewer sessions were identified as higher churn risk segments.

8. Business Recommendations

Develop targeted engagement strategies for younger users. Implement behavioral notifications to re-engage inactive users. Introduce personalised dashboards and feature recommendations. Monitor churn indicators such as declining session frequency.

9. Conclusion

This project demonstrates how user engagement analytics can provide meaningful insights into customer behaviour and platform performance. The interactive Power BI dashboard allows stakeholders to explore engagement trends dynamically and make data-driven decisions that improve user retention and platform optimisation.